radiate. The poor physicists, not knowing what to believe, might be put out of business.)

The new, meshed laws had to be consistent with general relativity's laws of curved spacetime in the absence of quantum fields and with the laws of quantum fields in the absence of spacetime curvature. This and the demand for a perfect mesh, analogous to the demand that the rows and columns of a crossword puzzle mesh perfectly, turned out to determine the form of the new laws almost[11] completely. If the laws could be meshed consistently at all (and they must be, if the physicists' approach to understanding the Universe makes any sense), then they could be meshed only in the manner described by the new, agreed-upon laws of quantum fields in curved spacetime.

The requirement that the laws of physics mesh consistently is often used as a tool in the search for new laws. However, this consistency requirement has rarely exhibited such great power as here, in the arena of quantum fields in curved spacetime. For example, when Einstein was developing his laws of general relativity (Chapter 2), considerations of consistency could not and did not tell him his starting premise, that gravity is due to a curvature of spacetime; this starting premise came largely from Einstein's intuition. However, with this premise in hand, the requirement that the new general relativistic laws mesh consistently with Newton's laws of gravity when gravity is weak, and with the laws of special relativity when there is no gravity at all, determined the forms of the new laws almost uniquely; for example, it was the key to Einstein's discovery of his field equation.

In September 1975,I returned to Moscow for my fifth visit, bearing a bottle of White Horse scotch for Zel'dovich. To my surprise, I discovered that, although all the Western experts by now had agreed that Hawking was right and black holes can evaporate, nobody in Moscow believed Hawking's calculations or conclusions. Although several confirmations of Hawking's claims, derived by new, completely different methods, had been published during 1974 and 1975, those confirmations had had little impact in the U.S.S.R. Why? Because Zel'dovich and Starobinsky, the greatest Soviet experts, were disbelievers: They continued to maintain that, after a radiating black hole has lost all its spin, it must stop radiating, and it therefore cannot evaporate completely. I argued with Zel'dovich and Starobinsky, to no